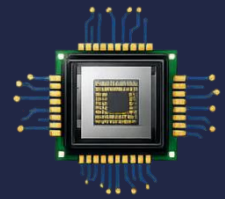
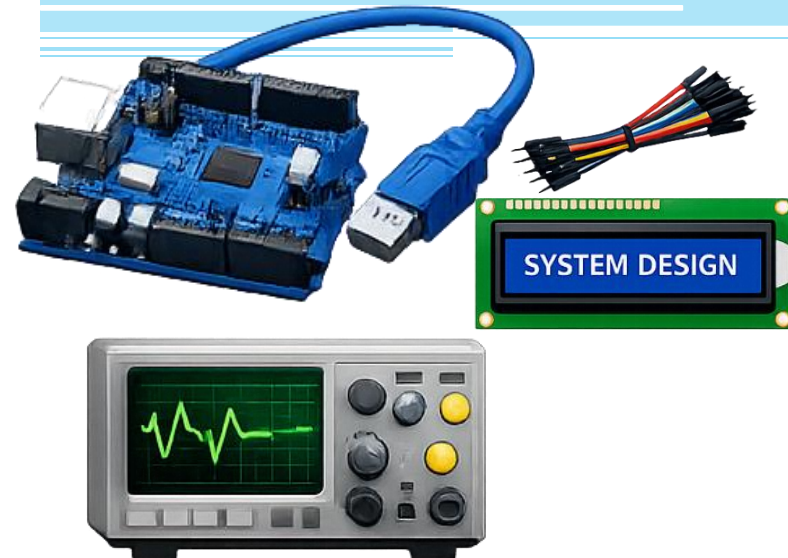


CS-301 Microprocessor- Based System Design



Lecture 2: IO Pins, Pin Functions, Programmable Peripheral Interfaces of AVR Controller

Course Teacher: Syeda Ramish Fatima



A Brief History

- AVR Core:
 - In 1996 two students at the **Norwegian Institute of Technology** (NTH) **Alf-Egil Bogen** and **Vegard Wollan** proposed the architecture of AVR Controllers
 - In 1996 **ATMEL** launch the first AVR controller
 - AVR
 - **A**LF + **V**egard + **R**ISC = **AVR**

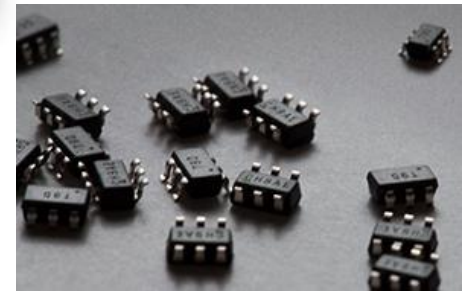
AVR Controllers

- The 3 different Flavors of AVR
- The AVR microcontrollers are divided into three groups:
 1. tinyAVR
 2. MegaAVR
 3. XmegaAVR

AVR Family of Microcontrollers

- **AVR TINY**

- 8 Bit
- Small and powerful
- General Purpose + Customized
- 0.5 – 8KB Flash
- 6-32 pins
- Up to 20MHz



- **AVR MEGA**

- 8 Bit
- More peripherals and options
- General Purpose + Customized
- 4-256KB Flash
- 28-100-pins
- Up to 20 MHz



AVR Family of Microcontrollers

- **AVR XMEGA**
 - 8/16 Bit
 - Xtreme performance
 - General Purpose
 - 16-384KB Flash
 - 44-100 pins
 - Up to 32



ATmega162

- The ATmega162 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture.
- By executing powerful instructions in a single clock cycle, the ATmega162 achieves throughputs approaching 1 MIPS per MHz allowing the system designer to optimize power consumption versus processing speed.

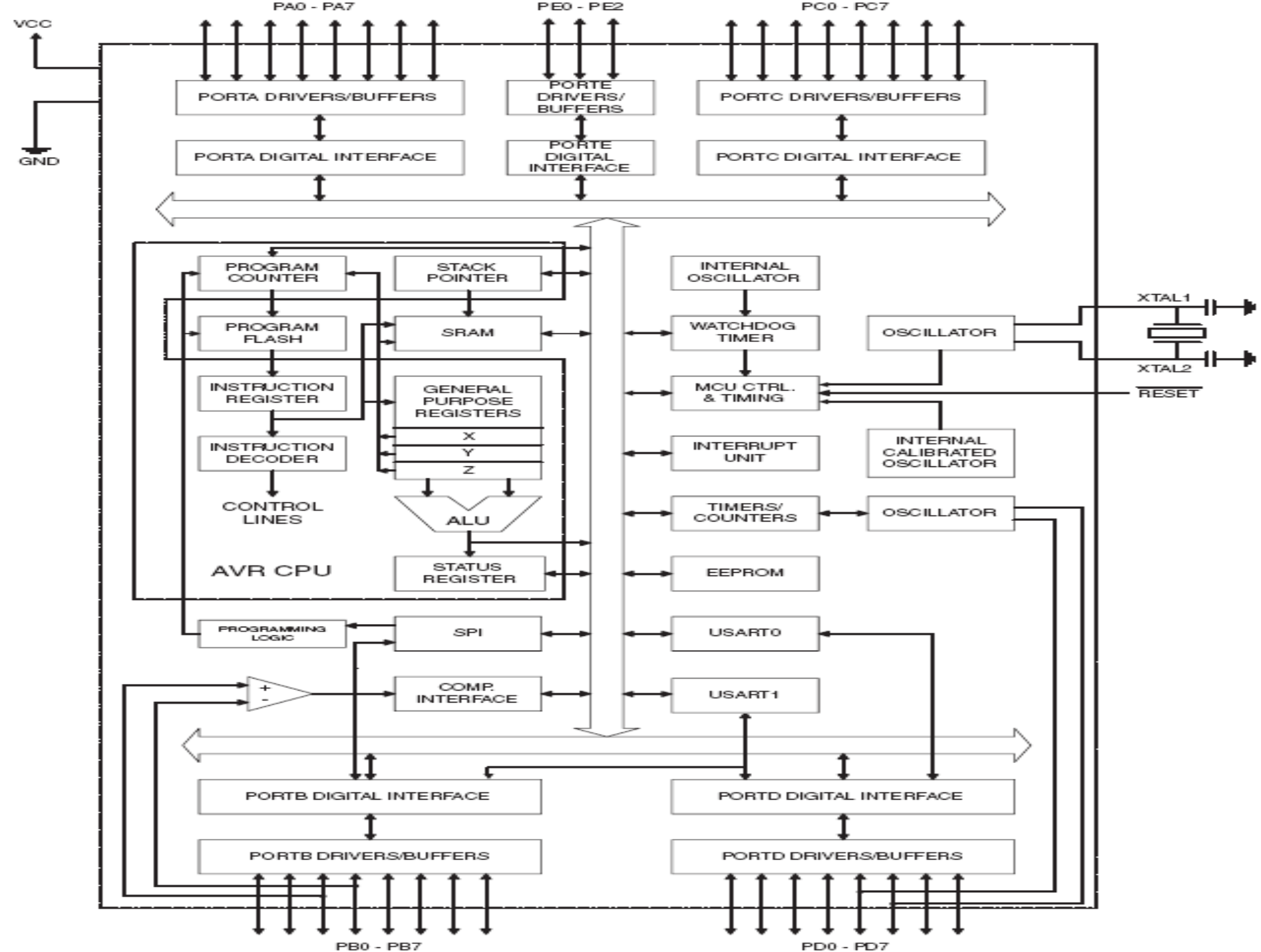
ATmega162

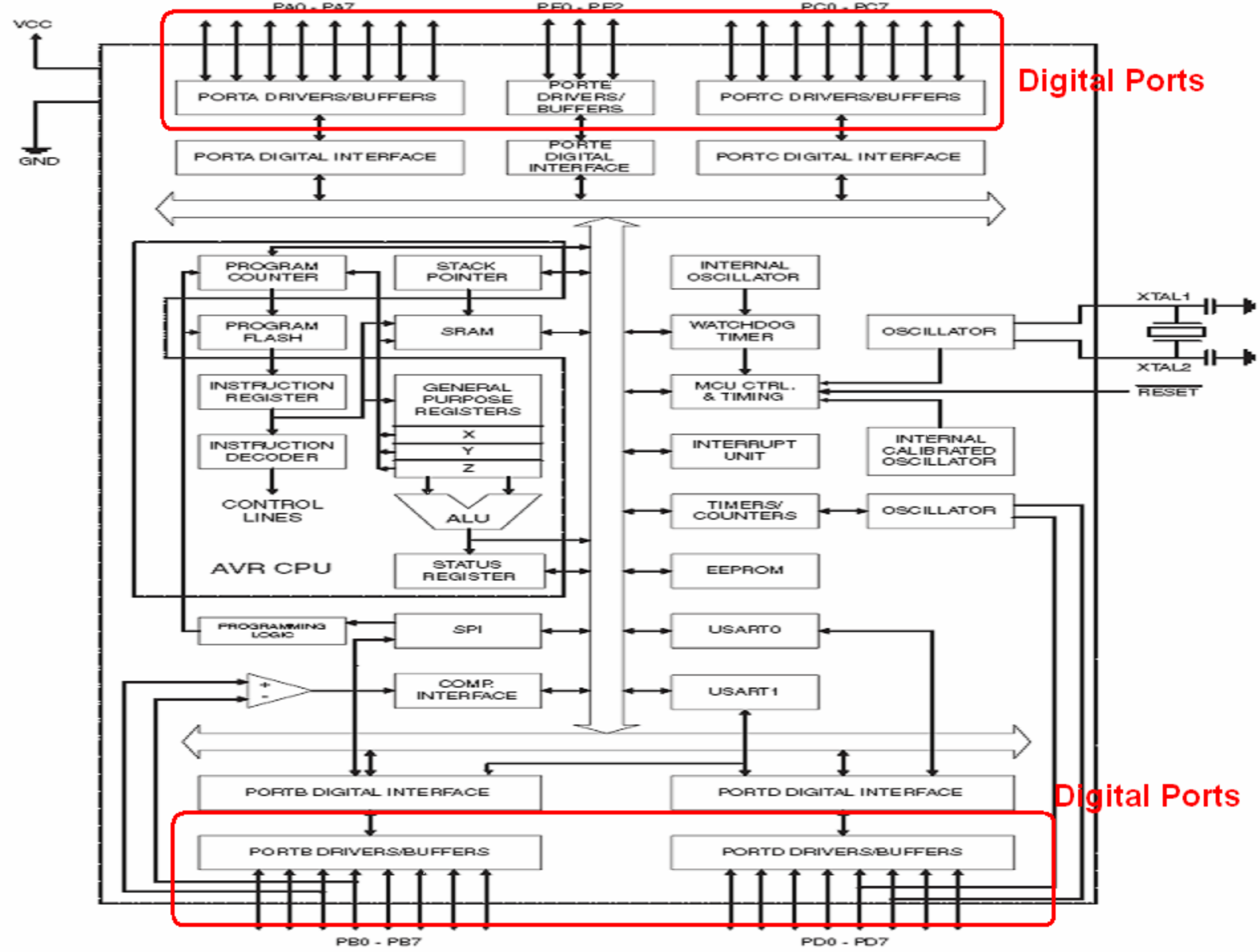
- The ATmega162 is a low-power CMOS 8-bit microcontroller based on the AVR enhanced RISC architecture.
- By executing powerful instructions in a single clock cycle, the ATmega162 achieves throughputs approaching 1 MIPS per MHz allowing the system designer to optimize power consumption versus processing speed.
- [..\Atmel-2513-8-bit-AVR-Microcontroller-ATmega162_Datasheet.pdf](#)

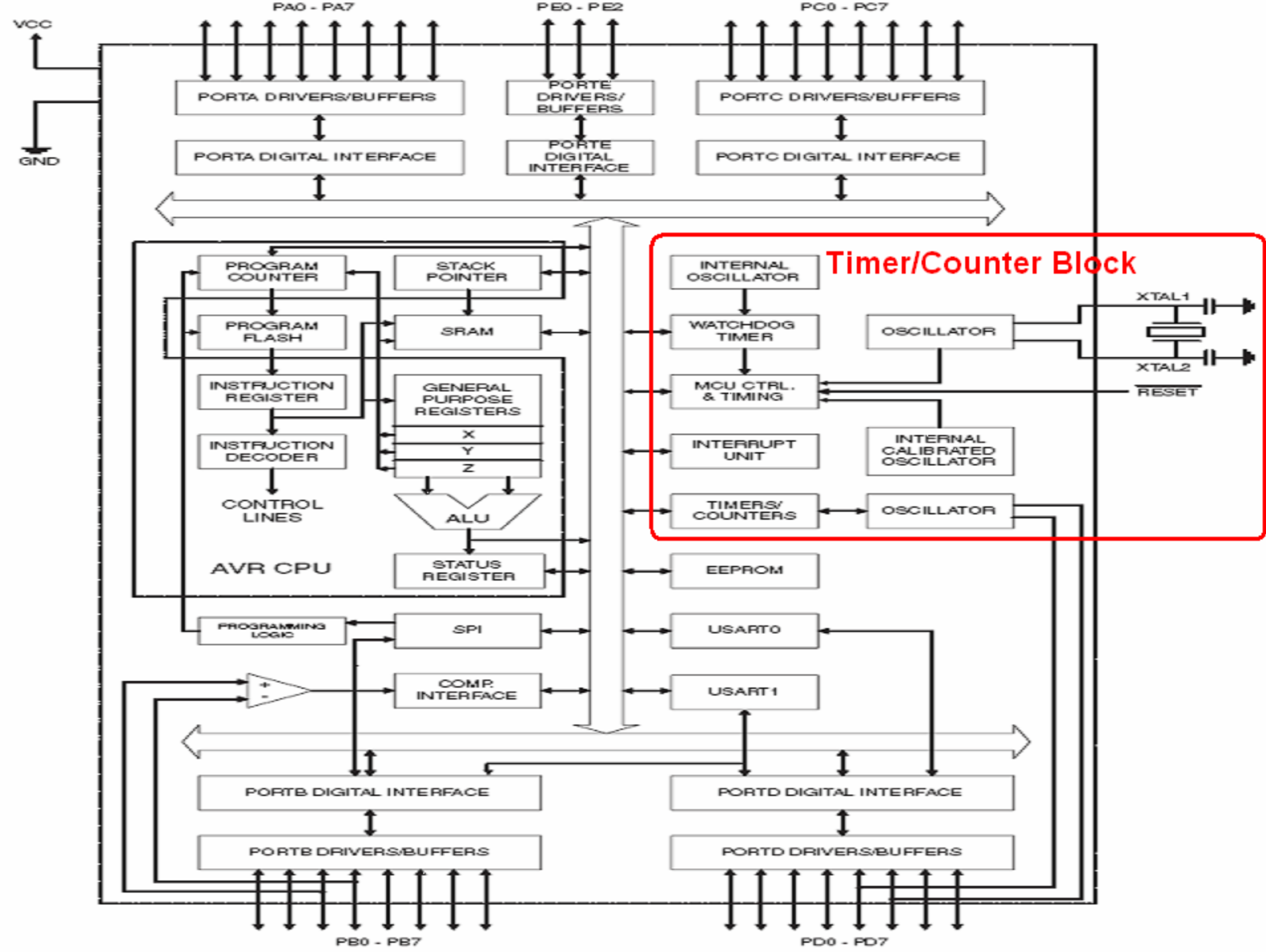
AVR ATMEGA-162

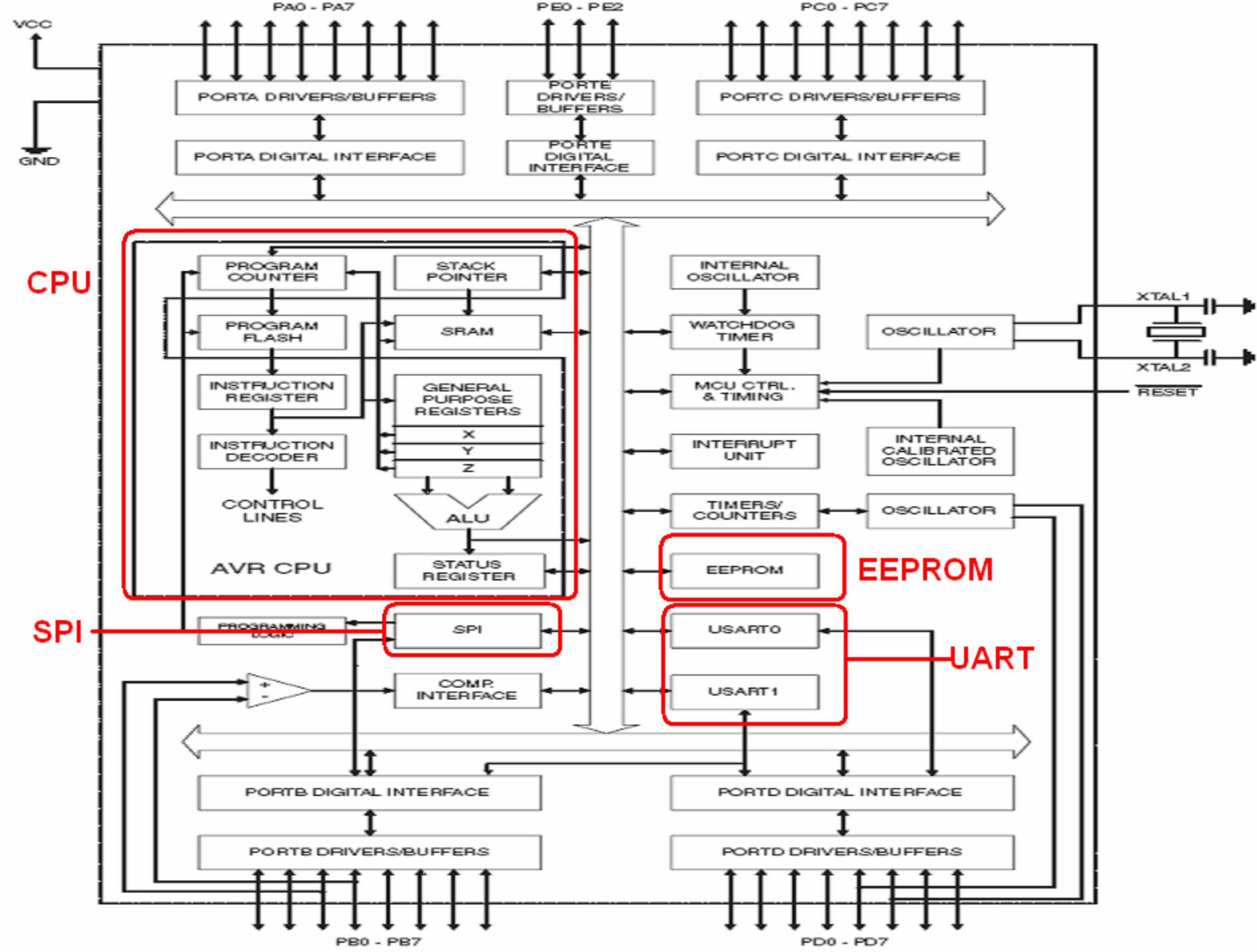
Salient Feature:

- 16K bytes of In-System Programmable Flash
- 512 bytes EEPROM, 1K bytes SRAM
- 35 general purpose I/O lines,
- 32 general purpose working registers,
- Four flexible Timer/Counters with compare modes,
- internal and external interrupts,
- two serial programmable USARTs,
- a programmable Watchdog Timer with Internal Oscillator,
- An SPI port,
- Five software selectable power saving modes.

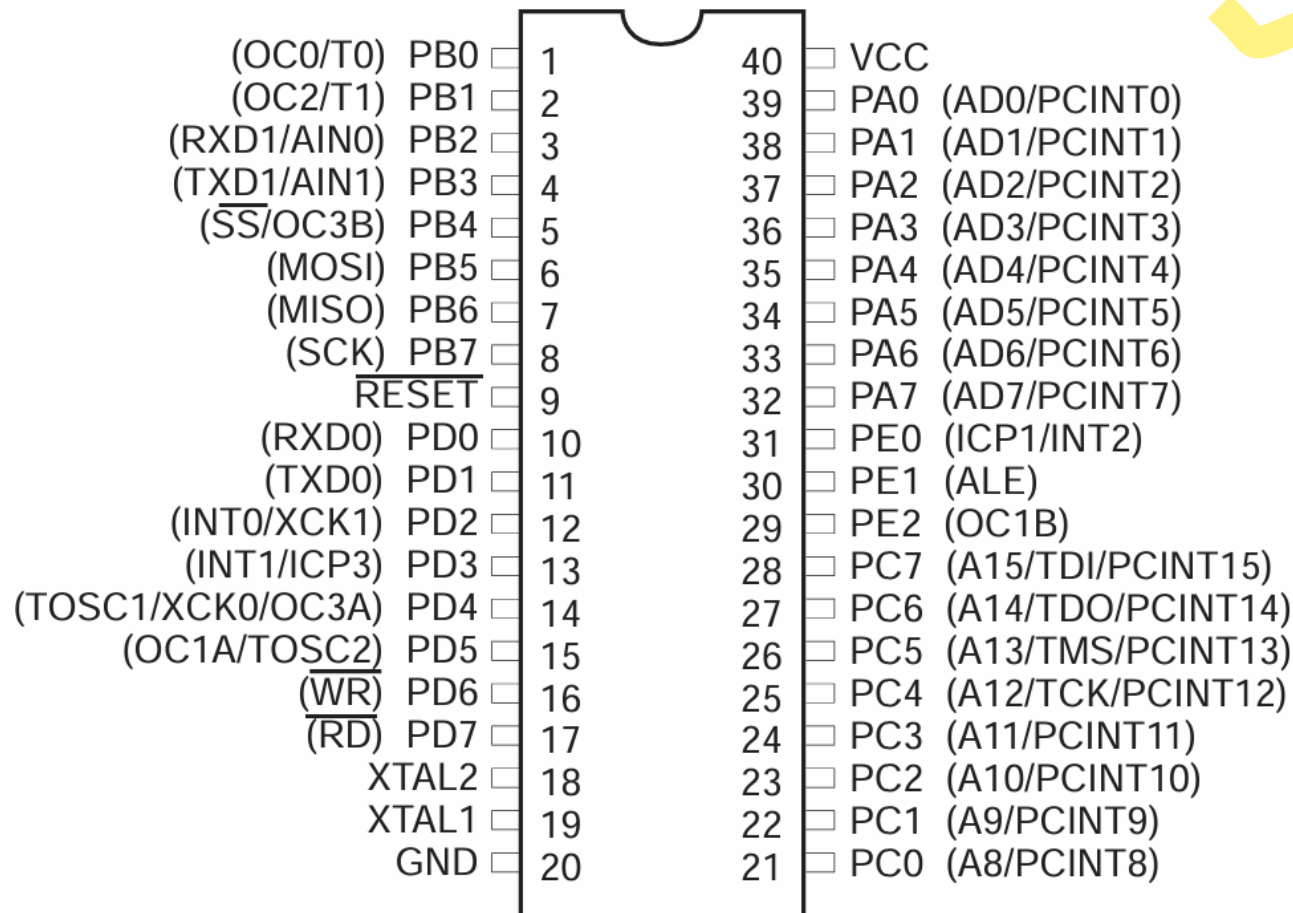








Pin Diagram of ATmega162

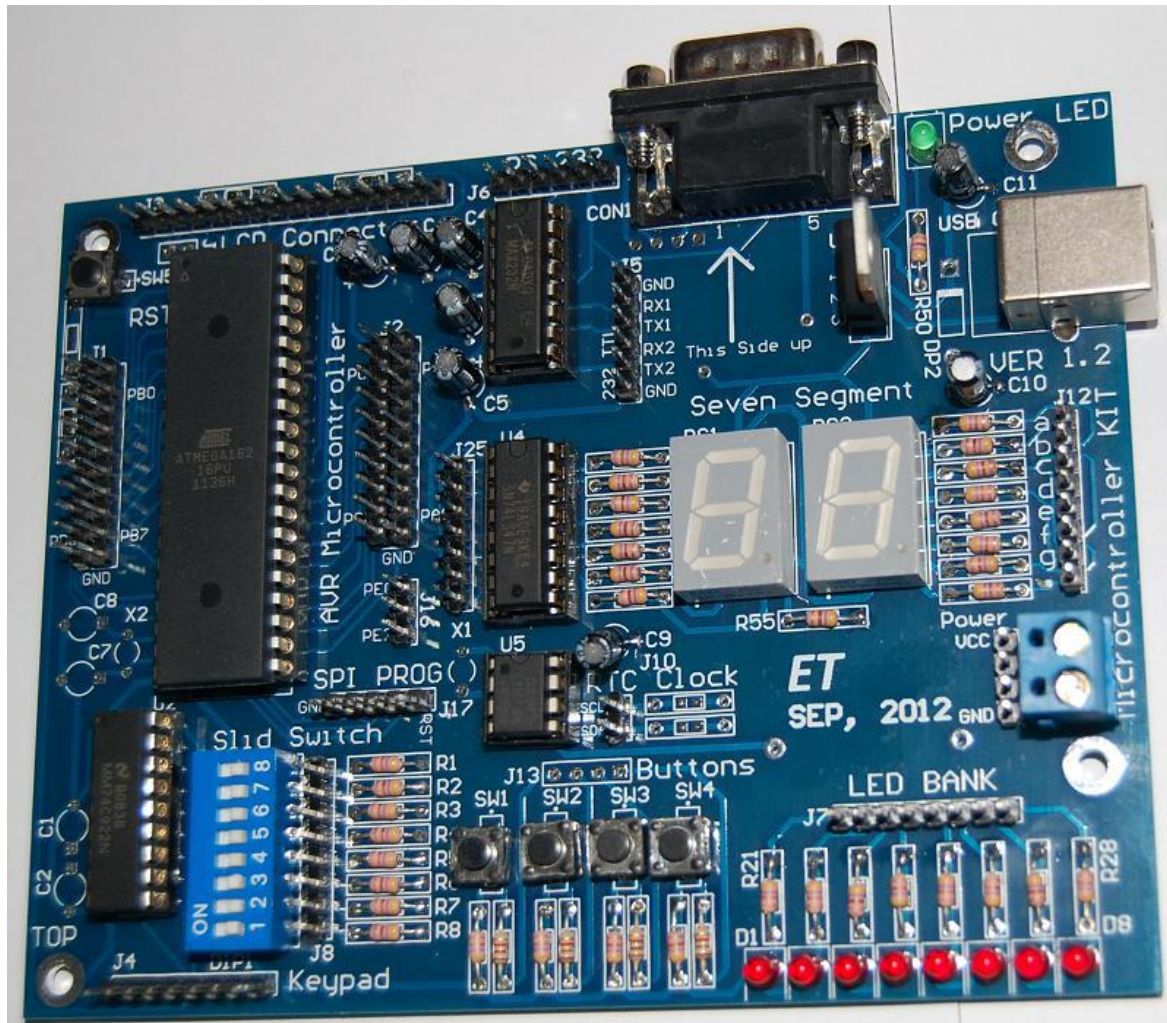


PDIP

PORTS

(OC0/T0) PB0	1	40	VCC
(OC2/T1) PB1	2	39	PA0 (AD0/PCINT0)
(RXD1/AIN0) PB2	3	38	PA1 (AD1/PCINT1)
(TXD1/AIN1) PB3	4	37	PA2 (AD2/PCINT2)
(<u>SS</u> /OC3B) PB4	5	36	PA3 (AD3/PCINT3)
(MOSI) PB5	6	35	PA4 (AD4/PCINT4)
(MISO) PB6	7	34	PA5 (AD5/PCINT5)
(SCK) PB7	8	33	PA6 (AD6/PCINT6)
<u>RESET</u>	9	32	PA7 (AD7/PCINT7)
(RXD0) PD0	10	31	PE0 (ICP1/INT2)
(TXD0) PD1	11	30	PE1 (ALE)
(INT0/XCK1) PD2	12	29	PE2 (OC1B)
(INT1/ICP3) PD3	13	28	PC7 (A15/TDI/PCINT15)
(TOSC1/XCK0/OC3A) PD4	14	27	PC6 (A14/TDO/PCINT14)
(OC1A/TOSC2) PD5	15	26	PC5 (A13/TMS/PCINT13)
(<u>WR</u>) PD6	16	25	PC4 (A12/TCK/PCINT12)
(<u>RD</u>) PD7	17	24	PC3 (A11/PCINT11)
XTAL2	18	23	PC2 (A10/PCINT10)
XTAL1	19	22	PC1 (A9/PCINT9)
GND	20	21	PC0 (A8/PCINT8)

AVR Kit



I/O Ports of ATmega162

- All AVR ports have true Read-Modify-Write functionality when used as general digital I/O ports.
- This means that the direction of one port pin can be changed without unintentionally changing the direction of any other pin with the SBI and CBI instructions.
- The same applies when changing drive value (if configured as output) or enabling/disabling of pull-up resistors (if configured as input).
- Each output buffer has symmetrical drive characteristics with both high sink and source capability.
- The pin driver is strong enough to drive LED displays directly.
- All port pins have individually selectable pull-up resistors with a supply-voltage invariant resistance.
- All I/O pins have protection diodes to both VCC and Ground as indicated.

IO Ports of ATmega162

- All registers and bit references in this section are written in general form.
- A lower case “x” represents the numbering letter for the port, and a lower case “n” represents the bit number.
- However, when using the register or bit defines in a program, the precise form must be used.
 - For example, PORTB3 for bit no. 3 in Port B, here documented generally as PORTxn.

Bit	7	6	5	4	3	2	1	0
	PORTA7	PORTA6	PORTA5	PORTA4	PORTA3	PORTA2	PORTA1	PORTA0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
	DDA7	DDA6	DDA5	DDA4	DDA3	DDA2	DDA1	DDA0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

[illegible]

Port C Data Register – PORTC

Bit	7	6	5	4	3	2	1	0
	PORTC7	PORTC6	PORTC5	PORTC4	PORTC3	PORTC2	PORTC1	PORTC0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Port C Data Direction Register – DDRC

Bit	7	6	5	4	3	2	1	0	
	DDC7	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	DDRC
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port C Input Pins

Address – PINC

[illegible]

Port D Data Register – PORTD

Bit	7	6	5	4	3	2	1	0
	PORTD7	PORTD6	PORTD5	PORTD4	PORTD3	PORTD2	PORTD1	PORTD0
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Initial Value	0	0	0	0	0	0	0	0

Port D Data Direction Register – DDRD

Bit	7	6	5	4	3	2	1	0	
	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port D Input Pins

Address – PIND

[illegible]

Register Description for I/O Ports



Port E Data Register – PORTE

Bit	7	6	5	4	3	2	1	0	
	–	–	–	–	–	PORTE2	PORTE1	PORTE0	PORTE
Read/Write	R	R	R	R	R	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port E Data Direction Register – DDRE

Bit	7	6	5	4	3	2	1	0	
	–	–	–	–	–	DDE2	DDE1	DDE0	DDRE
Read/Write	R	R	R	R	R	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Port E Input Pins Address – PINE

Bit	7	6	5	4	3	2	1	0	
	–	–	–	–	–	PINE2	PINE1	PINE0	PINE
Read/Write	R	R	R	R	R	R	R	R	
Initial Value	0	0	0	0	0	N/A	N/A	N/A	

Configuring the Pins

- Each port pin consists of three register bits: DDxn, PORTxn, and PINxn.
- As shown in “Register Description for I/O-Ports” on page 82(Datasheet), the DDxn bits are accessed at the DDRx I/O address, the PORTxn bits at the PORTx I/O address, and the PINxn bits at the PINx I/O address.

Configuring the Pins

- The DDxn bit in the DDRx Register selects the direction of this pin.
 - If DDxn is written logic one, Pxn is configured as an output pin.
 - If DDxn is written logic zero, Pxn is configured as an input pin.
 - If PORTxn is written logic one when the pin is configured as an input pin, the pull-up resistor is activated.
 - To switch the pull-up resistor off, PORTxn has to be written logic zero or the pin has to be configured as an output pin.
 - The port pins are tri-stated when a reset condition becomes active, even if no clocks are running.
 - If PORTxn is written logic one when the pin is configured as an output pin, the port pin is driven high (one).
 - If PORTxn is written logic zero when the pin is configured as an output pin, the port pin is driven low (zero).

Configuring the Pins

Table 27. Port Pin Configurations

DDxn	PORTxn	PUD (in SFIOR)	I/O	Pull-up	Comment
0	0	X	Input	No	Tri-state (Hi-Z)
0	1	0	Input	Yes	Pxn will source current if ext. pulled low.
0	1	1	Input	No	Tri-state (Hi-Z)
1	0	X	Output	No	Output Low (Sink)
1	1	X	Output	No	Output High (Source)

Configuring the Pins

Bit	7	6	5	4	3	2	1	0	
	TSM	XMBK	XMM2	XMM1	XMM0	PUD	PSR2	PSR310	SFIOR
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

Bit 2 – PUD: Pull-up Disable in SFIOR(Special Function IO Register)

When this bit is written to one, the pull-ups in the I/O ports are disabled even if the DDxn and PORTxn Registers are configured to enable the pull-ups ({DDxn, PORTxn} = 0b01).

See “Configuring the Pin” on page 64 for more details about this feature

Alternate Functions of PORT A

Table 29. Port A Pins Alternate Functions

Port Pin	Alternate Function
PA7	AD7 (External memory interface address and data bit 7) PCINT7 (Pin Change INTerrupt 7)
PA6	AD6 (External memory interface address and data bit 6) PCINT6 (Pin Change INTerrupt 6)
PA5	AD5 (External memory interface address and data bit 5) PCINT5 (Pin Change INTerrupt 5)
PA4	AD4 (External memory interface address and data bit 4) PCINT4 (Pin Change INTerrupt 4)
PA3	AD3 (External memory interface address and data bit 3) PCINT3 (Pin Change INTerrupt 3)
PA2	AD2 (External memory interface address and data bit 2) PCINT2 (Pin Change INTerrupt 2)
PA1	AD1 (External memory interface address and data bit 1) PCINT1 (Pin Change INTerrupt 1)
PA0	AD0 (External memory interface address and data bit 0) PCINT0 (Pin Change INTerrupt 0)

Alternate Functions of PORT B

Table 32. Port B Pins Alternate Functions

Port Pin	Alternate Functions
PB7	SCK (SPI Bus Serial Clock)
PB6	MISO (SPI Bus Master Input/Slave Output)
PB5	MOSI (SPI Bus Master Output/Slave Input)
PB4	$\overline{SS}$ (SPI Slave Select Input) OC3B (Timer/Counter3 Output Compare Match Output)
PB3	AIN1 (Analog Comparator Negative Input) TXD1 (USART1 Output Pin)
PB2	AIN0 (Analog Comparator Positive Input) RXD1 (USART1 Input Pin)
PB1	T1 (Timer/Counter1 External Counter Input) OC2 (Timer/Counter2 Output Compare Match Output)
PB0	T0 (Timer/Counter0 External Counter Input) OC0 (Timer/Counter0 Output Compare Match Output) clk _{I/O} (Divided System Clock)

Alternate Functions of PORT C

Table 35. Port C Pins Alternate Functions

Port Pin	Alternate Function
PC7	A15 (External memory interface address bit 15) TDI (JTAG Test Data Input) PCINT15 (Pin Change INTerrupt 15)
PC6	A14 (External memory interface address bit 14) TDO (JTAG Test Data Output) PCINT14 (Pin Change INTerrupt 14)
PC5	A13 (External memory interface address bit 13) TMS (JTAG Test Mode Select) PCINT13 (Pin Change INTerrupt 13)
PC4	A12 (External memory interface address bit 12) TCK (JTAG Test Clock) PCINT12 (Pin Change INTerrupt 12)
PC3	A11 (External memory interface address bit 11) PCINT11 (Pin Change INTerrupt 11)
PC2	A10 (External memory interface address bit 10) PCINT10 (Pin Change INTerrupt 10)
PC1	A9 (External memory interface address bit 9) PCINT9 (Pin Change INTerrupt 9)
PC0	A8 (External memory interface address bit 8) PCINT8 (Pin Change INTerrupt 8)

Alternate Functions of PORT D

Table 38. Port D Pins Alternate Functions

Port Pin	Alternate Function
PD7	$\overline{RD}$ (Read strobe to external memory)
PD6	$\overline{WR}$ (Write strobe to external memory)
PD5	TOSC2 (Timer Oscillator Pin 2) OC1A (Timer/Counter1 Output Compare A Match Output)
PD4	TOSC1 (Timer Oscillator Pin 1) XCK0 (USART0 External Clock Input/Output) OC3A (Timer/Counter3 Output Compare A Match Output)
PD3	INT1 (External Interrupt 1 Input) ICP3 (Timer/Counter3 Input Capture Pin)
PD2	INT0 (External Interrupt 0 Input) XCK1 (USART1 External Clock Input/Output)
PD1	TXD0 (USART0 Output Pin)
PD0	RXD0 (USART0 Input Pin)

Alternate Functions of PORT E

Table 41. Port E Pins Alternate Functions

Port Pin	Alternate Function
PE2	OC1B (Timer/Counter1 Output CompareB Match Output)
PE1	ALE (Address Latch Enable to external memory)
PE0	ICP1 (Timer/Counter1 Input Capture Pin) INT2 (External Interrupt 2 Input)